

Bitcoin: Blockchain Security and Its Connection to the Simple Random walk

Hanaa Zaita
Amal El moussaoui
Alessia Van der auwermeulen

January 09, 2026

Abstract

This project explores the fundamental concepts and mechanisms of Bitcoin and blockchain technology. We study the security of Proof-of-Work blockchains through a probabilistic analysis of double-spend attacks.

The mining competition is modeled as a biased random walk with an absorbing state, leading a closed-form expression for the attack success probability. Monte Carlo simulations confirm the theoretical results and verify the exponential decay of attack probability with the number of confirmations.

Contents

1	Introduction	2
2	Network	2
3	Analytical Probabilistic Modeling of Double-Spend Attacks	3
3.1	Mining Power Distribution and Block Creation	3
3.2	Markov chain and Random Walk Model of the Blockchain	3
3.3	Gambler's Ruin and Attack Success Probability	5
3.4	Nakamoto Poisson Model	6
3.5	Monte Carlo Simulation for Model Verification	7
4	Monte Carlo Simulations and Limit Theorems	8
5	Conclusion	10

1 Introduction

With the rapid growth of digital payment systems, ensuring secure and reliable transactions has become a major challenge. Unlike physical cash, digital assets can be duplicated at negligible cost, giving rise to the double-spending problem, where the same digital unit may be spent more than once.

Blockchain-based cryptocurrencies, most notably Bitcoin, were introduced as a decentralized solution to this issue. Proposed in 2008 by Satoshi Nakamoto, Bitcoin was designed as a peer-to-peer electronic cash system relying on cryptographic proof rather than trust. At its core lies the blockchain, a publicly shared ledger in which transactions are recorded in successive blocks linked together in a verifiable manner, without the need for a central authority.

The security of the blockchain relies on Proof of Work, a mechanism in which miners compete to add new blocks by solving cryptographic puzzles. Because block creation is inherently random, the evolution of the blockchain results from a stochastic competition between miners. While this mechanism is robust, it does not make the system absolutely secure.

In particular, a double-spending attack may occur if an attacker manages to build a competing blockchain and eventually overtake the honest one, reversing a previously accepted transaction. The success of such an attack depends on the attacker's relative computational power and on the number of confirmations waited before accepting a transaction.

This competition can naturally be modeled using probabilistic tools. In this report, we analyze double-spending attacks by modeling the difference between the honest chain and the attacker's chain as a stochastic process. We combine theoretical results based on random walks and Markov chains with Monte Carlo simulations in Python to quantify the security of the Bitcoin protocol.

2 Network

On the Bitcoin network, all users are notified of every transaction. These transactions are collected by miners and grouped into blocks. Miners compete to solve a computationally difficult Proof-of-Work problem in order to add a new block to the blockchain.

Each block contains transactions, a timestamp, the hash of the previous block, a block number, and a nonce. Miners repeatedly modify this nonce until the block meets a predefined difficulty condition, making block validation and propagation times inherently random.

Due to network delays, nodes may temporarily observe different versions of the blockchain and eventually adopt the longest valid chain. In a double-spending attack, an attacker privately mines an alternative chain in an attempt to overtake the honest one. This competition can be modeled as a biased random walk, forming the basis of our probabilistic analysis.

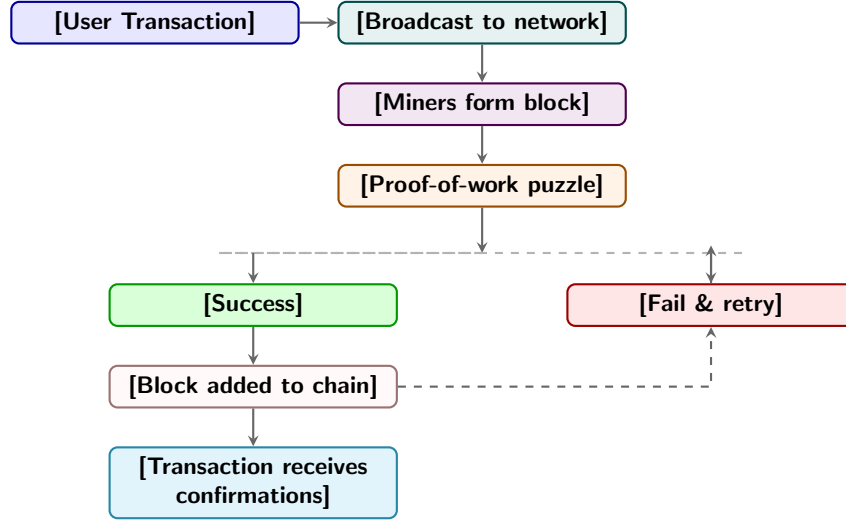


Figure 1: Blockchain Network Transaction Flow through Proof-of-Work

3 Analytical Probabilistic Modeling of Double-Spend Attacks

3.1 Mining Power Distribution and Block Creation

We consider a blockchain network composed of two competing entities: the honest miners and a potential attacker. Let $q \in (0, 1)$ denote the fraction of total computational power controlled by the attacker, and let $p = 1 - q$ denote the fraction controlled by the honest network.

Block creation is governed by the Proof-of-Work mechanism, which can be modeled as a sequence of independent trials. At each block discovery event, either the honest miners or the attacker successfully mine a block. This outcome can be represented as a Bernoulli trial with success probabilities:

$$P(\text{honest mines a block}) = p, \quad P(\text{attacker mines a block}) = q.$$

This probabilistic abstraction captures the randomness inherent in the mining process and forms the basis of the subsequent stochastic modeling.

3.2 Markov chain and Random Walk Model of the Blockchain

Let X_n denote the difference in length between the honest chain and the attacker's chain after n blocks:

$$X_n = (\text{length of honest chain}) - (\text{length of attacker chain}).$$

- Initially, the honest chain is ahead by z confirmations:

$$X_0 = z.$$

- At each block discovery, the process evolves as a one-dimensional biased random walk:

$$X_{n+1} = \begin{cases} X_n + 1, & \text{with probability } p, \\ X_n - 1, & \text{with probability } q, \end{cases}$$

where $p + q = 1$.

This random walk can be represented as a "discrete-time Markov chain" with state space $\mathcal{S} = \mathbb{Z}$ and transition matrix P given by

$$P = \begin{pmatrix} \ddots & \ddots & & & \\ \ddots & 0 & p & & \\ & q & 0 & p & \\ & & q & 0 & \ddots \\ & & & \ddots & \ddots \end{pmatrix}.$$

Here, the (i, j) -th entry of P represents

$$P_{ij} = \mathbb{P}[X_{n+1} = j \mid X_n = i].$$

The drift of the walk is

$$\mathbb{E}[X_{n+1} - X_n] = p - q > 0,$$

indicating that the honest chain grows faster on average. The absorbing states (if any) correspond to the attacker catching up completely or failing definitively.

We now illustrate the stochastic dynamics of the mining race through a sample random walk, followed by the attack success probabilities under varying numbers of confirmations and attacker mining power.

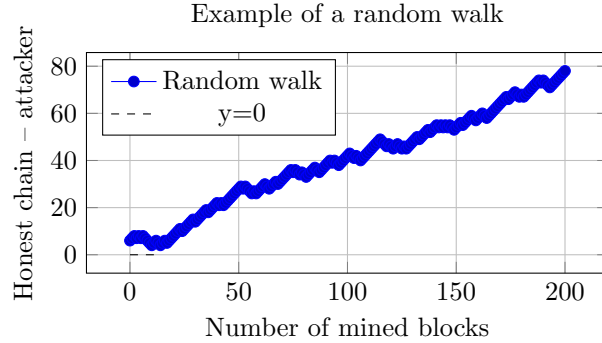


Figure 2: Trajectory of a mining race

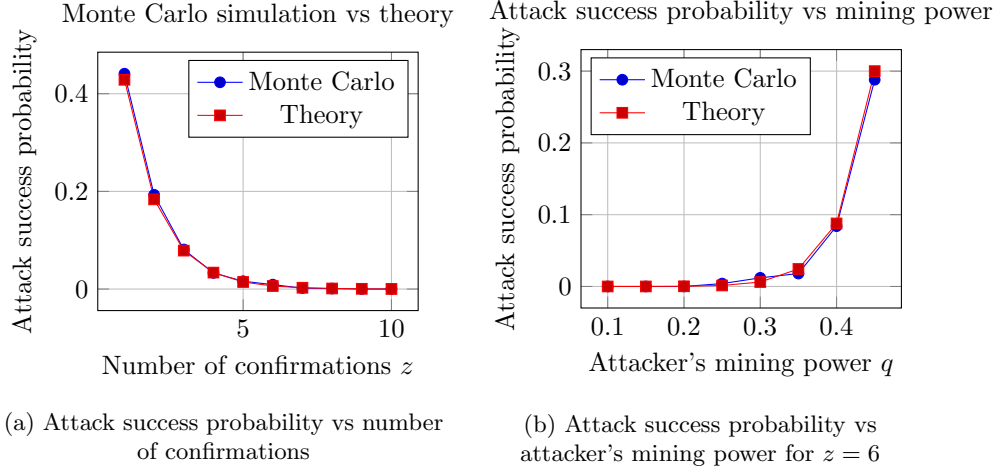


Figure 3: Comparison of attack success probability: (a) vs number of confirmations, (b) vs mining power

Figure 2 illustrates the random walk, showing the evolution of the difference between the honest chain and the attacker's chain as blocks are mined. Even when $p \neq q$, stochastic fluctuations can occasionally bring this difference to zero. Despite these temporary declines, the honest chain generally maintains its lead, reflecting the random yet resilient behavior of the blockchain.

Figure 3(a) presents the attack success probability versus the number of confirmations. The plot indicates that the likelihood of a successful attack decreases exponentially as the number of confirmations increases. Monte Carlo simulations closely match the theoretical predictions, confirming that higher number of confirmations significantly reduces the risk of attack.

Figure 3(b) illustrates the attack probability versus attacker's mining power. The graph shows that the success probability grows with the attacker's fraction of total hash power, but remains low as long as the attacker controls less than 50% of the mining power. These results highlight the security of the blockchain when the majority of miners are honest.

Overall, these figures collectively demonstrate that the blockchain is robust against attacks, with security improving both with the number of confirmations and the presence of an honest majority in the network.

3.3 Gambler's Ruin and Attack Success Probability

The classical **Gambler's Ruin problem** provides a convenient framework to compute the probability that a biased random walk, starting from $X_0 = z$, eventually reaches zero (i.e., the attacker catches up with the honest chain). For a one-dimensional random walk with transition probabilities p and q , the

probability that the attacker ever overtakes the honest chain is given by

$$P_{\text{success}}(z) = \begin{cases} \left(\frac{q}{p}\right)^z, & \text{if } q < p, \\ 1, & \text{if } q \geq p, \end{cases}$$

where z is the number of confirmations by which the honest chain initially leads.

3.4 Nakamoto Poisson Model

We refine the Gambler's Ruin model by taking into account the randomness of block mining times. Under the Proof-of-Work mechanism, block discoveries occur at random times and can be modeled by a **Poisson process**.

Let $p = 1 - q$ denote the fraction of honest mining power, q the attacker's fraction, and z the number of confirmations required by the merchant.

During the time needed for the honest network to mine z blocks, the number of blocks mined by the attacker is modeled as a Poisson random variable K with parameter

$$\lambda = z \frac{q}{p}.$$

Conditionally on $K = k$, the attacker remains behind by $z - k$ blocks. The probability that the attacker eventually catches up from this deficit is given by the Gambler's Ruin result:

$$Q_{z-k} = \begin{cases} 1, & \text{if } q \geq p, \\ \left(\frac{q}{p}\right)^{z-k}, & \text{if } q < p. \end{cases}$$

Therefore, the overall probability of a successful double-spend attack is

$$Q(z) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} Q_{z-k}.$$

Simulation Procedure (Poisson Model)

To simulate this model, the procedure is as follows:

1. Fix z and q , and compute $\lambda = z \frac{q}{p}$.
2. Sample $K \sim \text{Poisson}(\lambda)$.
3. If $K \geq z$, the attacker immediately catches up.
4. Otherwise, the attacker succeeds with probability $\left(\frac{q}{p}\right)^{z-K}$.
5. Repeat the procedure many times to estimate $Q(z)$.

The theoretical probability $Q(z)$ given by Nakamoto's Poisson model can be plotted as a function of the number of confirmations z , for several values of $q < 0.5$. This illustrates how the success probability of a double-spending attack decreases rapidly with the number of confirmations.

3.5 Monte Carlo Simulation for Model Verification

To verify the accuracy of the theoretical Poisson model, we perform a Monte Carlo simulation of the block mining process. For each set of parameters z (number of confirmations) and q (attacker's fraction of mining power), we proceed as follows:

1. Compute $\lambda = z \frac{q}{p}$ with $p = 1 - q$, representing the expected number of blocks mined by the attacker while the honest miners mine z blocks.
2. Sample $K \sim \text{Poisson}(\lambda)$ to obtain the number of blocks the attacker actually mines in this interval.
3. If $K \geq z$, the attacker immediately catches up, resulting in a successful double-spend.
4. Otherwise, the attacker remains behind by $z - K$ blocks and succeeds with probability $(q/p)^{z-K}$ according to the Gambler's Ruin result.
5. Repeat the above steps for a large number of trials N , and estimate the success probability $Q(z)$ as the fraction of trials in which the attacker succeeds.

This Monte Carlo approach allows us to capture the stochastic variability in block discovery times and serves as an empirical validation of the theoretical formula

$$Q(z) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} Q_{z-k}.$$

In practice, the simulated probabilities converge closely to the theoretical values, confirming the correctness of the Poisson model in representing the dynamics of double-spending attacks under the Proof-of-Work mechanism.

The following graphs illustrate the attack success probabilities under two theoretical models: the Gambler's Ruin and the Nakamoto Poisson model.

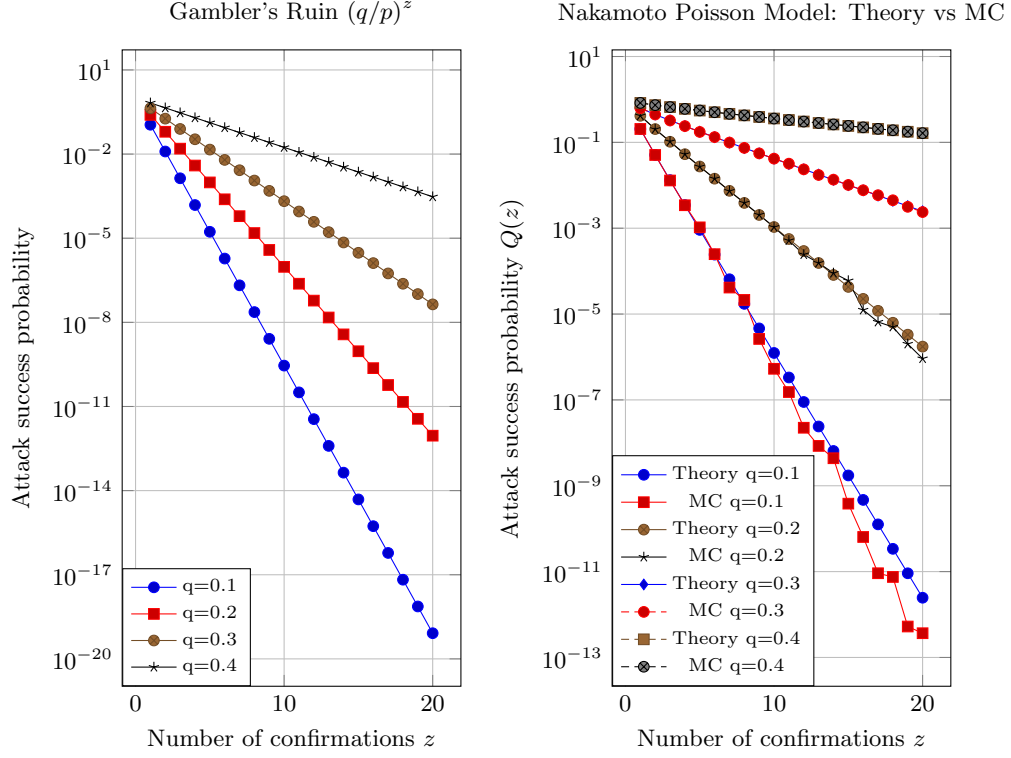


Figure 4: Attack success probabilities: Blockchain simulations and Poisson models

Figure 4 represents two complementary views of blockchain attack probabilities. The first plot (Gambler’s Ruin) illustrates that the probability of a successful attack decays exponentially with the number of confirmations z for different attacker mining powers q .

The second plot (Nakamoto Poisson model) compares theoretical predictions with Monte Carlo simulations, which closely follow the theory and validate the Poisson approximation. Although the attack probability increases with the attacker’s mining power, it remains low as long as the attacker controls less than 50% of the network.

These results demonstrate how both models capture the stochastic nature of blockchain security and confirm the effectiveness of confirmations in reducing attack risks.

4 Monte Carlo Simulations and Limit Theorems

This section examines Monte Carlo estimators for attack probability, showing how the Law of Large Numbers (LLN) ensures convergence to the true value as the number of simulations increases. For each N , the estimator $\hat{p} = \frac{1}{N} \sum_{i=1}^N X_i$ is obtained by averaging independent Bernoulli trials, where X_i indicates a

successful attack. The 95% confidence interval, $\hat{p} \pm 1.96 \sqrt{\hat{p}(1 - \hat{p})/N}$, illustrates the estimator's convergence, while repeating the simulations multiple times for a fixed N shows that $\hat{p} \sim \mathcal{N}(p, p(1 - p)/N)$, as predicted by the Central Limit Theorem (CLT). These results highlight both the stochastic variability of the estimator and the predictive power of classical limit theorems.

The following figures present the corresponding Monte Carlo estimates, their distribution, and confidence intervals.

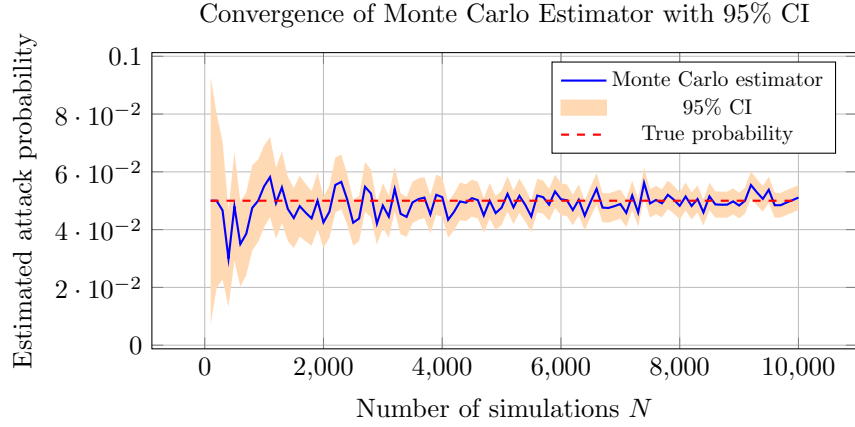


Figure 5: Convergence of the Monte Carlo estimator with 95% confidence interval.

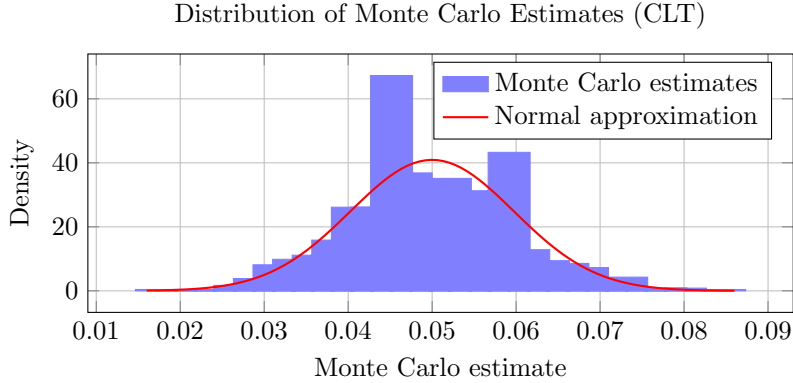


Figure 6: Histogram of Monte Carlo estimates and their normal approximation.

Figure 5 shows the Monte Carlo estimator of the attack probability as a function of the number of simulations N . The estimate (blue) fluctuates for small N but stabilizes near the true probability (red dashed), while the 95% confidence interval (shaded orange) narrows, illustrating the Law of Large Numbers.

Figure 6 displays the distribution of Monte Carlo estimates for a fixed number of simulations $N = 500$ over multiple runs. The histogram shows the empirical distribution of the estimates (blue bars), while the red curve represents the theoretical normal approximation predicted by the Central Limit Theorem. The close match between the histogram and the normal curve confirms that the estimator is approximately normally distributed around the true probability, with variance decreasing as N increases. This highlights the stochastic behavior of the Monte Carlo method and the Central Limit Theorem.

5 Conclusion

In this report, we used Monte Carlo simulations to study blockchain mining and the probability of successful double-spending attacks. The numerical results were in close agreement with theoretical models based on Gambler’s Ruin and the Nakamoto Poisson framework, confirming the impact of confirmation depth and attacker mining power on blockchain security.

Our results show that when the attacker controls less than half of the total mining power, the probability of a successful attack decreases exponentially with the number of confirmations, making such attacks extremely unlikely in practice. This highlights that blockchain security is not absolute but probabilistic, relying on both the distribution of mining power and the number of confirmations used.

Overall, this study demonstrates that combining probabilistic modeling with Monte Carlo simulations provides an effective framework for quantifying blockchain security and understanding the conditions under which the Bitcoin protocol remains robust.

References

- [1] S. Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System*, 2008. Available at: <https://bitcoin.org/bitcoin.pdf>
- [2] Y. Laguel, *Blockchain – Mathematical Analysis*, Lecture notes, Université Côte d’Azur.
- [3] R. Diel and S. Rubenthaler, *Méthodes de Monte Carlo et chaînes de Markov*, Master 1 lecture notes, Université Côte d’Azur, 2025.
- [4] A. Pinar Ozisik and B. N. Levine, *An Explanation of Nakamoto’s Analysis of Double-Spend Attacks*, arXiv:1701.03977, 2017.